Music4All: A New Music Database and Its Applications

Igor André Pegoraro Santana, Fabio Pinhelli, Juliano Donini, Leonardo Catharin, Rafael Biazus Mangolin,
Yandre Maldonado e Gomes da Costa, Valéria Delisandra Feltrim, Marcos Aurélio Domingues
Department of Informatics (DIN) - State University of Maringá (UEM) - Maringá, PR, Brazil
{*pg400816, pg400815, pg400595, pg400809, pg400811, ymgcosta, vdfeltrim, madomingues*}*@uem.br*

***Abstract*—One of the goals of the music information retrieval (MIR) community is to research new methods and create new systems that can efficiently and effectively retrieve and recommend songs from large databases of music content. Despite the volume of research in the area, there is a lack of music databases to support these works, i.e. databases that comply with some quite desirable requirements for the development of researches, such as: a huge amount of music pieces, the audio signal availability and a great diversity of audio attributes. In order to contribute to the MIR community, we present Music4All, a new music database which contains metadata, tags, genre information, 30-seconds audio clips, lyrics, and so on. Additionally, we also exemplify some MIR tasks that may benefit from our database and compare it with other databases proposed in the literature. To request the Music4All database, send an e-mail to contact4music4all@gmail.com.**

***Keywords*—Music Information Retrieval (MIR), Music Database, MIR Applications**

I. Introduction

Music goes beyond what we know as an element for fun or distraction, as it plays an important role on education and development of the human being in society [1]. It is a social phenomenon capable of maintaining traditional roles in different societies, even over the centuries. Through music, children can change how they think, act and establish relationships with friends transposing the idea of a simple song when raising its level to an element of social formation [2].

Music information retrieval (MIR) is a research area that has been rapidly gaining attention due to the widespread digital distribution of music. Examples of such distribution are commercial applications like Spotify[1], Apple Music[2] and Pandora[3]. In 2017, the commercialization of digital media presented a rise in revenue of 41.1%, while the revenue from the physical media dropped 5.4%. This growth represents a profit of 9 billion dollars for the sector, the highest recorded profit until now [3]. These numbers emphasize the relevance of the area and the many opportunities in it.

One of the goals of MIR is to research new methods and create new systems that can efficiently and effectively retrieve and recommend songs from large databases of music content. However, there is a lack of music databases to support this kind of research. Although several different databases have already been presented in the literature, there is still a lack of databases that comply with some quite desirable requirements for the development of researches, such as: a huge amount of music pieces, the audio signal availability and a great diversity of audio attributes.

Aiming at contributing to the MIR community, in this paper we present Music4All, a new music database which contains metadata, tags, genre information, audio clips, lyrics, and so on which are essentials for the research and development of new digital music systems. We describe how we collected, filtered, and organized the data, and also demonstrate how the database can be used in three traditional MIR tasks. The Music4All database, including 30-seconds clips, lyrics and other attributes, is available under request[4].

[1]https://www.spotify.com
[2]https://www.apple.com/music
[3]https://www.pandora.com
[4]Send an e-mail to contact4music4all@gmail.com for more information.

The remaining of this paper is organized as follows. In Section II we describe the process that generated the Music4All database. Three MIR tasks that can be explored based on Music4All are presented in Section III. In Section IV we compare our database with other music databases in the literature. Finally, conclusions and future work are presented in Section V.

II. The Database

Music4all was designed to be a useful database in different domains of research. Therefore, it needed to gather different types of information related to the songs. To achieve this purpose, the development process was split into two phases, called the user phase and the music phase. In the user phase we gather information about the listening history of users, and in the song phase we collected data about the songs (i.e. audio, lyrics, etc).

Figure 1 presents a summary of the entire database development process. The steps of the two phases are represented by rectangles and the data sources from which the data were retrieved are represented by out-of-rectangles database icons. The steps that contain a filter icon on the top-right are steps in which data was removed (filtered) from the original database. Filtering steps were performed to prevent songs with missing information from being kept in the database.

The user phase was initiated by sampling a random last.fm user as a starting point. From this user, we gathered all of its friends through the last.fm API and recursively obtained the circle of friends, excluding duplicates. We collected 40,940 users in this step. For all retrieved users, we collected their listening histories ranging from January 1st, 2019 00:00 to March 20th, 2019 23:59. We delimited this range of dates to obtain recent, popular and well-known songs.

In the following step, we anonymized the database by discarding any personal information and metadata collected through the last.fm API[5], including the relationship between users. We generated a new randomly identifier for each user.

Finally, to remove outliers (users that did not have much songs) and produce a denser database, we used a threshold to filter the users and keep only the ones that had at least 1,000 plays in their listening histories. Users that did not meet this threshold were discarded, resulting in 15,602 users and 320,073 songs at the end of the user phase.

The song phase consisted of collecting data from songs, which included tags, lyrics, genres, audio, and metadata. We retrieved this data from different sources and merged it into the database. Again, to avoid keeping songs with missing information, we discarded the ones for which we were unable to retrieve all necessary data. Tags for each song were retrieved by a last.fm crawler that accessed each song's page. We decided to use a crawler because it can return more tags than the API. From a total of 320,073 songs, only 306,010 had one or more tags.

To retrieve genre information for each song, we used the genre names available at Every Noise At Once[6], an application that aims to

[5]https://www.last.fm/api
[6]http://everynoise.com

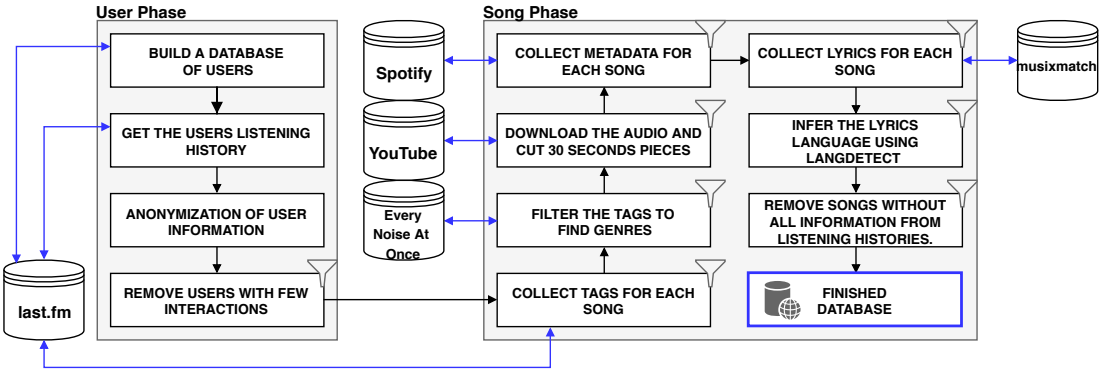


Fig. 1: Summary of the development of the database.

classify musical genres based on the genre-shaped distinctions made by Spotify. From the 306,010 songs that had tags associated with, we filtered out songs that had no genre names, resulting in 266,837 songs with genres.

In the following step, we obtained the audios from the songs by downloading them in mp3 format from YouTube[7]. To obtain the YouTube link, we used the same crawler used for collecting tags, as most of the songs from last.fm had YouTube links assigned to them. We were able to download audio for 218,429 out of 266,837 songs.

To comply with fair use requirements and avoid copyright infringement, each song was cut into a 30-seconds clip using the ffmpeg[8] library. The audio clipping was done by finding the mean point between the start and the end of the song, and then clipping 15 seconds before and after the mean point. Songs for which the audio did not meet the 30 seconds threshold were discarded. Eight songs were discarded in this step, resulting in 218,421 songs.

Most of the songs had metadata/attributes associated with it in the Spotify API. To collect this metadata, we used the API search method to associate the artist and song name to a Spotify ID. After retrieving the Spotify ID, the metadata was collected by using the API endpoint. A summary of the metadata is presented in Table I. Songs that were not found by the Spotify API search were discarded. The final amount of songs that had metadata associated with it was 136,390.

To retrieve the songs' lyrics, we crawled the Musixmatch[9] application. We were able to retrieve the lyrics for 111,924 songs. Since the language in which the lyrics were written may be an important information for processing it, we used the tool langdetect[10] to infer the lyrics' language. We chose to keep in the database only the songs for which the tool detected the language with a probability higher than 90%, so we ended up with 109,269 songs.

As a final step, we filtered out from the users' listening histories songs that were discarded at some point in the song phase. Therefore, the final version of the database contains 15,602 anonymous users, their listening histories, and 109,269 songs represented by their audio clips, lyrics, and 16 other metadata/attributes.

To better organize the database, we clustered the attributes into different files, which contain the id of the song in a column and the attributes in the other columns. Table II presents these files' names and the attributes contained in it.

Statistics regarding the listening histories from users and the distribution of tags and genres are presented in Figure 2. As showed in Figure 2a, a user has an average of 200 unique songs in its listening history. Figure 2b depicts the distribution of tags per the songs, showing that the majority of songs have two tags associated with them, and that the number of tags associated with a song ranges from 1 to 8. Figure 2c shows the distribution of genres per song and, as can be observed, most songs have only a few genres associated with them.

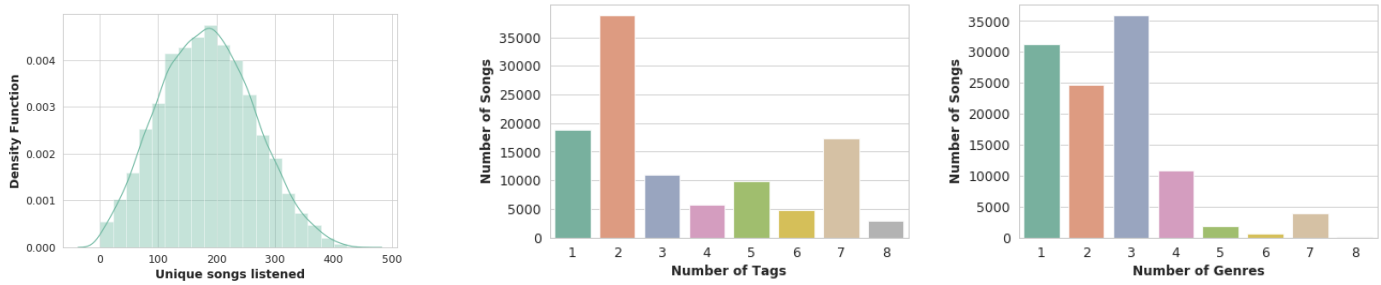
TABLE I: Music4All database metadata/attributes.

Attribute	Description
id	Unique 16 characters identifier for the song in the database.
artist	Name of the artist that published the song in last.fm. There are 16,269 unique artists in the database.
song	Name of the song.
lang	Language assigned to the lyrics by the tool langdetect. There are 46 unique languages in the database.
spotify_id	Song identifier in the Spotify application.
popularity	Integer value ranging from 0 to 100 representing how popular a song is on Spotify. This value is based on the total number of plays the song has and how recent those plays are.
album_name	Name of the album that the song is in. There are 38,363 different albums in the database.
release	Year in which the song was released.
danceability	Real value ranging from 0.0 to 1.0 representing how suitable the song is for dancing. This value is based on a combination of musical elements, provided by the Spotify API.
energy	Real value ranging from 0.0 to 1.0 provide by the Spotify API that is a perceptual measure of intensity and activity.
key	Overall key of the song, using standard Pitch Class notation, provided by the Spotify API.
mode	Binary value provided by the Spotify API corresponding to the modality of the song, where major is represented by 1 and minor by 0.
valence	Real value ranging from 0.0 to 1.0 provided by the Spotify API that measures how positive a song is.
tempo	Speed or pace of the song, measured in beats per minute (BPM), provided by the Spotify API.
genres	List of genres tags associated with the song. There are 853 unique genre tags in the database.
tags	User-given tags from the last.fm application, with 19,541 unique tags.

TABLE II: File names and a brief description of its contents.

Filename	Description
id_genres.csv	File containing the id of the song and its list of genres, separated by a comma.
id_information.csv	File containing the id of the song along with basic information about it, namely artist, song name and album name.
id_lang.csv	File containing the id of the song and its lyrics language.
id_metadata.csv	File containing the id of the song, its attributes obtained through the Spotify official API, including its Spotify id.
id_tags.csv	File containing the id of the song and the tags associated with it, separated by a comma.
listening_history.csv	File containing the listening history for each user in the database.

[7]https://www.youtube.com
[8]https://ffmpeg.org
[9]https://www.musixmatch.com
[10]https://pypi.org/project/langdetect



(a) Amount of unique songs a user listened to. (b) Distribution of tags per song. (c) Distribution of genre tags per song.

Fig. 2: Statistics for the database.

III. MIR Applications using Music4All

To demonstrate how useful our database can be for the MIR community, we developed applications based on it in three traditional areas of MIR research, namely music recommendation, genre classification, and mood classification. These applications and the results obtained for Music4All are described in the following subsections.

A. Music Recommendation

The main goal of a Music Recommender System is to propose to a given user interesting songs to listen to, as well as unknown artists and their tracks, based on the personal preference from the user [4]. One of the applications in which our database can be applied, in this research area, is suggesting playlists to the user, and this subsection focuses on this application.

As the database contains the listening history for each user, and the timestamp in which each song was listened by the user; the listening history for each user was rearranged into sessions. To split into sessions, songs that were listened by the same user 60 minutes apart from each other were placed in different sessions.

To represent the songs and capture the intrinsic features of the sequence in which they were listened by a given user, we used the two models proposed by [5]: music2vec and session-music2vec. Those models were built using as a basis the word2vec model [6]. The difference between those models is that music2vec infers the user's preference from the entirety of its listening history, with the intent of learning the general preference from music, while session-music2vec infers the contextual preference from the current session only (i.e. the context of the listening), focusing on the current interest of the user. Both preferences are calculated by the mean average of the vector representations of the songs in the listening history and the current session. These preferences are used by the recommendation algorithms.

The session-based algorithms used to suggest playlists were the ones proposed by [5]: Music2vec-TopN (M-TN), SessionMusic2vec-TopN (SM-TN), Context-Session-Music2vec-TopN(CSM-TN) and Context-Session-Music2vec-UserKNN (CSM-UK).

The M-TN algorithm determines the preference between a user and a song by the cosine similarity between the user's preference and the song embeddings obtained by music2vec. The SM-TN is similar to the M-TN algorithm, but it uses the contextual preference and the song embeddings obtained by session-music2vec.

CSM-TN implements a combination of the previous two algorithms. The preference is obtained by adding both the similarities from the M-TN and the SM-TN algorithms. As for the CSM-UK algorithm, it is an adaptation of UserKNN [7] with the learned embeddings from music2vec and session-music2vec models.

To evaluate the algorithms we followed the methodology proposed by [5]. We carried out a crossfold evaluation in which the users were split into 5 folds. The training data for each fold had all the listening histories from the training users in addition to the first half of the listening histories of the test users. The test data held the second half of the listening histories of the test users.

We used Precision, Recall, and F-measure as evaluation metrics. In [8], Precision is described as the fraction of recommended items that are relevant to the user, and Recall is described as the fraction of relevant items that are recommended. F-measure can then be interpreted as the harmonic mean between the Precision and Recall.

As can be seen in Figure 3, the music2vec model did not learn useful general preferences for users in the M-TN algorithm, reaching an F-measure value of only 0.037. Although the low values for the general preferences, the session-music2vec performed better with respect to the contextual preference, with the F-measure going up to 0.126 in the SM-TN algorithm.

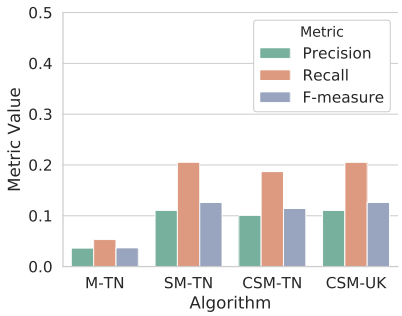


Fig. 3: Results for the algorithms proposed by [5].

The M-TN algorithm, that combines the two preferences, presented a lower F-measure than the SM-TN algorithm. As we can see in Figure 3, not only the music2vec did not do a great job in learning the general preference, but when combined with the contextual preference, it reduces the F-measure to 0.114. The CSM-UK algorithm presented results similar to SM-TN algorithm.

This application shows that Music4All can be used in music recommendation research by using the listening history data. As the results for the contextual preferences were higher than the general preferences, it indicates that context-aware recommender systems might benefit from the database as well.

B. Genre Classification

Automatic music genre classification is an important task in MIR, and several works approach this task with different techniques and data sources [9], [10]. By knowing that a song contains different data, such as lyrics, audio, video clip, and metadata, and that each one of those has different characteristics of the same song, some data will be more

suitable to a certain task than to others. When performing music genre classification, audio-based approaches receives a significant amount of attention from the MIR community [11], showing good results and presenting the audio as an important data source for this task. Due to this fact, we also used the audio from the songs in this application.

Several features can be used in audio-based music genre classification, from traditional features like Mel-frequency cepstral coefficients (MFCCs) [12] and Statistical Spectrum Descriptors (SSDs) [13] (extracted from the raw audio) to representation learning features (e.g. Convolutional Neural Networks (CNNs)). Like in other machine learning tasks, deep learning approaches such as CNNs have overcome the results of traditional techniques and achieved state-of-the-art results in music genre classification as well.

Most of the work in the music genre classification area approaches a multi-class scenario, where each music has only one genre. But, in a real-world application, music usually has more than one genre tag associated to it, characterizing a multi-label classification problem. In a multi-label problem, each instance is assigned to one or more labels [14]. In this work, we performed multi-label genre classification using CNN with the spectrograms generated from the audio clips. The CNN architecture used was the Inception_v3 [15], a robust CNN model that has achieved good results in various classification tasks.

The experimental protocol was performed with 5-fold cross-validation by using artist filtering [16] with a minimum of 100 music pieces per genre in each fold, resulting in a subset with 65 genres tags and 103,568 songs. To run the experiments, we used the 30-seconds audio clip provided by the database and extracted its spectrogram representation using SoX library [17].

We have used the Inception_v3 [15] with its pre-trained weights using the ImageNet Dataset [18]. After the convolutional layers, which gives us the feature maps, it has two dense layers with 2,048 nodes that use ReLU [19] as the activation function. Then, we have used the sigmoid as the activation function in the classification layer because its a multi-label classification problem. This layer was set to have 65 nodes to match the number of genres in our subset.

The results obtained with the Inception_v3 are presented in Table III. In this table, if we take into account metrics that consider only correct predictions, the results are 0.2786 for the Accuracy and 0.3703 for the F-measure. If we consider the True Negative predictions (where the model correctly predicts that the song does not have a particular genre), the results increase considerably, as for the 0.9029 value obtained by the AUC-ROC metric.

TABLE III: Incept_v3 predictions for multi-label genre classification.

Metric	Accuracy	F-Measure	AUC-ROC
Mean	0.2786	0.3703	0.9029
Standard Deviation	0.0137	0.0143	0.0050

In the application of multi-label genre classification, we could not achieve state-of-the-art results, but further research could be done using the Music4All database. It is worth noticing that, for obvious reasons, this database was evaluated here for the first time in this application. So, it is not possible to make a fair comparison to other results on genre classification already described in the literature.

C. Mood Classification

Applications such as retrieving music by genre or emotion have stood out in the MIR area. In particular, Music Emotion Recognition (MER) has received increasing attention from the scientific community, as emotion is a fundamental attribute from a song [20].

Several works in the literature propose automatic methods for MER. The most common way to approach this application is to categorize songs using specific emotions labels, such as happy, sad, calm, angry [21]–[23]. Other way to approach the problem is by representing emotion, or aspects related to it, as continuous values [24], [25]. The songs in our database contain continuous values of valence (polarity) and energy (arousal), which in turn can be used for emotion recognition, so we have adopted the second approach.

Although there is more information available in the database, it is fairly common to find works [22], [26], [27] that use only lyrics for mood prediction. Therefore, we used only lyrics on this application.

We used two Long Short-Term Memory (LSTM) model, one to infer valence values and one to infer energy values. Both are fairly simple LSTM model with three layers: an embedding layer, which learns a representation of the word in real-valued vector; a LSTM layer, which learns intrinsic features from the input data; and a dense layer, that maps the features learnt by the LSTM to a final prediction. The entire model can be seen in the Figure 4.

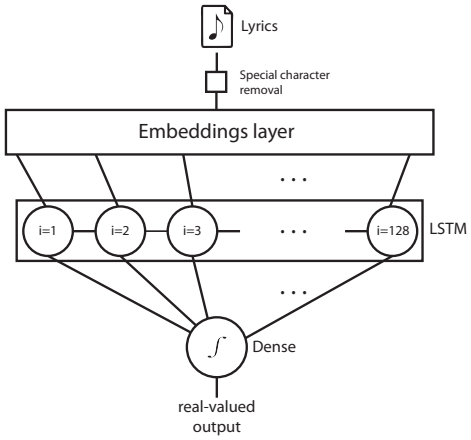


Fig. 4: LSTM model architecture.

The following parameters values were used in our model: the input sequence size was set to 500 words and the embedding layer output dimension value was set to 200 dimensions. The LSTM layer had 128 units, and the data was fed to the model with a batch size of 2.048 in 100 epochs. Finally, we used the sigmoid as activation function for the dense layer.

Experiments were performed with 5-fold cross-validation and we used Mean Absolute Error (MAE), Spearman Correlation and Pearson Correlation as evaluation metrics. Experimental results are presented in Table IV. The values are in the format $Means \mp Std$, where $Means$ is the average of 5 folds and Std is the respective Standard Deviation.

TABLE IV: Results for valence and energy.

Metrics	MAE	Spearman	Pearson
Valence	0.2102 ∓ 0.0027	0.3422 ∓ 0.0116	0.3426 ∓ 0.0108
Energy	0.1974 ∓ 0.0028	0.3497 ∓ 0.0105	0.3295 ∓ 0.0115

As can be observed, the results for valence and energy are similar. In both cases, MAE is close to 0.20 and the correlation coefficients are close to 0.34, indicating a weak positive correlation between real and predicted values. The standard deviation also indicates that there was little variance among the results of the folds.

The performance of the application is promising, but further work is required. Improvements would include optimizing the parameters of the LSTM model, evaluating other models and deep architectures, and combining different modalities, like text and audio.

IV. Related Works

In this section we describe some of the most important music databases used by the research community to support the development

of works on MIR. We briefly remark some historical facts about these databases, and other details about them, as the number of music pieces available, audio signal availability, number of class labels and so on. The databases are presented predominantly in chronological order.

RWC is one of the first databases presented to the MIR research community. It is a copyright-cleared music database, originally made available as a common foundation for the research development. RWC is the acronym for Real World Computing, and it was created by the RWC Music Database Sub-Working Group of the Real World Computing Partnership (RWCP) of Japan. Although the RWC database has shown to be quite useful for a while, it is undeniable that its small number of samples (i.e. only one hundred of music pieces) imposed limitations to address the new challenges which have emerged.

GTZAN is among the most traditional music databases used by the MIR research community for investigations on the genre classification task. This database was presented by Tzanetakis and Cook [28], who were also the first to treat music genre classification as a pattern recognition task. In addition to the database, the authors also presented baseline results, which encouraged other researchers to use the same database for comparison purposes. The database is balanced regarding the number of samples per class, there are one hundred music pieces classified in ten genres, namely Blues, Classical, Country, Disco, Hip Hop, Jazz, Metal, Popular, Reggae, and Rock.

Berenzweig et al. [29] presented the USPOP 2002 database, composed of 8,752 music pieces assigned to ten genres. The authors made available the full content of the songs under request, to ensure the use for research purposes.

Originally created to support the development of the ISMIR 2004 Audio Description Contest, ISMIR 2004 has become one of the most widely used databases for music genre classification because it was chosen as one of the databases to be explored in the genre classification task in several editions of the annual Music Information Retrieval Evaluation eXchange (MIREX), a contest that happens as part of the conference of the International Society for Music Information Retrieval (ISMIR). Although it was created and launched in 2004, it was described in details in the Technical Report of the Music Technology Group of the University Pompeu Fabra [30] in 2006. The database provides the full audio content of 1,458 music pieces, labeled according to one of the following six classes: Classical, Electronic, Jazz/Blues, Metal/Punk, Rock/Pop, and World.

Homburg et al. presented in [31] a new database, called HOMBURG set, created for MIR research. The database is composed of 1,886 music title assigned to nine different music genres, namely Blues, Electronic, Jazz, Pop, Rap/HipHop, Rock, Folk/Country, Alternative, and Funk/Soul. The use of the database remained limited because only 10 seconds of signal of each music was provided.

Gouyon et al. [32] performed experiments on a ballroom database collected from BallroomDancers.com[11]. The database is composed of 698 instances from eight dancing styles. The audio clips provided have 30-seconds duration.

The Computer Audition Lab 500 (CAL500) database is made up of 500 Western popular song excerpts of different lengths [33]. The excerpts were annotated with a set of 174 different tags.

Holzapfel and Stylianou [34] used a dance styles database that contains music clips assigned to six classes of dances commonly encountered in the island of Crete: Kalamatianos, Siganos, Maleviziotis, (fast) Pentozalis, Sousta and Chaniotis. Each class contains either fifteen or sixteen instrumental song excerpts of about ten seconds length. It is important to observe that, as in the case of the ballroom database previously described, this kind of classes are more suitable for investigation about "tempo induction" than "music genre", as evidenced in the "tempo induction" contest occurred as part of the annual ISMIR conference in 2004.

The 1517-artists database[12] is composed of 3,180 music titles assigned to nineteen labels. It was organized by Seyerlehner et al. [35], and the songs are available under request.

The Latin Music Database (LMD) was presented by Silla et al. [36] in 2008. The database has become one of the databases used in the genre contest in the annual ISMIR conference. The most important remark about LMD is that it is a database composed of Latin music genres, quite different from traditional western genres, typically provided in the other widely used databases. The database is composed of 3,227 samples from ten different genres, namely Axé, Pagode, Bachata, Bolero, Forró, Gaúcha, Merengue, Salsa, Sertanejo, and Tango. Despite of the new challenges provided by this database, its use is limited by the fact that the audio content itself was not made available, due to copyright restrictions.

The Magnatagatune database consists of 21,642 excerpts of length 30 seconds from 230 different artists collected by using the TagATune game [37]. Excerpts annotations are among a set of 188 tags.

The Million Songs Dataset (MSD) is the biggest database already created for MIR research purposes [38]. Despite this remarkable characteristic, the music content is not publicly available, therefore limiting its use. The database is composed of one million music titles, and it was launched in 2011 as a collaborative project between The Echo Nest and LabROSA[13], from the Columbia University.

Table V presents a summary of the characteristics of the databases described in this section.

TABLE V: Some of the most important music databases.

Database	Number of Instances	Number of Classes	Available Content	Authoriz. Requested	Songs Lenght
RWC [39]	100	10	Songs	Yes	Full
GTZAN [28]	1,000	10	Songs	No	Songs
USPOP 2002 [29]	8,752	10	Songs	Yes	Full
ISMIR 2004 [30]	1,458	6	Songs	No	Full
Homburg [31]	1,886	9	Songs	No	10 sec
Ballroom [32]	698	8	Songs	No	30 sec
CAL500 [33]	500	174	Songs	No	Full
Holzapfel [34]	N/A	6	Songs	Yes	Full
1517-artists [35]	3,180	19	Songs	Yes	Full
LMD [36]	3,227	10	Features	No	Full
Magnatagatune [37]	21,642	188	Songs	No	30 sec
MSD [38]	1,000,000	Variable	Features	No	Full
Music4All	109,269	Variable	Many	Yes	30 sec

As shown by the related works, more than one decade after the presentation of the pioneer works on MIR, this subject is still attracting attention from researchers. Although several different databases have already been presented, there is still a lack of databases that comply with some quite desirable requirements for the development of researches, such as: a huge amount of music pieces and the audio signal availability. Probably, it has been provoked by the limitations imposed to make the audio content itself available, due to copyright restrictions. By this way, in this work we try to contribute in this sense, introducing a novel music database that offers hundreds of thousands of music pieces assigned to metadata, tags, labels, lyrics, and so on. Moreover, it is important to remark that we make available a 30-seconds clip taken from the middle of the original music pieces, which is appropriate to the development of new research works.

[11]http://www.ballroomdancers.com

[12]http://www.seyerlehner.info

[13]http://millionsongdataset.com

V. Conclusion and Future Work

In this paper we presented Music4All, a new music database which contains metadata, tags, genre information, 30-seconds audio clips, lyrics, and so on. We started by describing in detail how the database was created. Additionally, we also presented some examples of MIR applications that may be developed by using our database. Finally, we compared it against other databases proposed in the literature.

Music4All was designed to be a database useful in distinct domains of research, and as future work we intent to explore it in many different applications. It is noteworthy that although we have exemplified the use of the database with applications that used only one of the types of information available, the database is also suitable for research involving multimodal applications. Thus, future work will also involve research focused on multimodal applications.

Acknowledgment

This study was financed in part by the Conselho Nacional de Desenvolvimento Científico e Tecnológico - (CNPq) - grants #403648/2016-5 and #156956/2018-7. We also acknowledge the support of NVIDIA Corporation with the donation of the Titan XP and V GPUs.

References

[1] A. H. Gregory, "The roles of music in society: The ethnomusicological perspective.," 1997.

[2] A. C. North, D. J. Hargreaves, and S. A. O'Neill, "The importance of music to adolescents," *British Journal of Educational Psychology*, vol. 70, no. 2, pp. 255–272, 2000.

[3] IFPI, "Global music report 2018 - annual state of the industry," tech. rep., International Federation of the Phonographic Industry, 2018.

[4] O. Celma, "Music Recommendation," in *Music Recommendation and Discovery*, Springer Berlin Heidelberg, 2010.

[5] D. Wang, S. Deng, and G. Xu *Information Retrieval Journal*, no. 2-3.

[6] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representation in vector space," *Comput. Research Repository*, 2013.

[7] P. Resnick, N. Iacovou, M. Suchak, P. Bergstrom, and J. Riedl, "GroupLens: An open architecture for collaborative filtering of netnews," in *Proc. of the ACM Conf. on Computer Supported Cooperative Work*, 1994.

[8] G. Shani and A. Gunawardana, "Evaluating Recommender Systems," in *DS '12: Recommender Systems Handbook*, pp. 257–298, Springer, 2011.

[9] S. Oramas, O. Nieto, F. Barbieri, and X. Serra, "Multi-label music genre classification from audio, text, and images using deep features," *arXiv preprint arXiv:1707.04916*, 2017.

[10] S. Oramas, F. Barbieri, O. Nieto, and X. Serra, "Multimodal deep learning for music genre classification," *Transactions of the International Society for Music Information Retrieval. 2018; 1 (1): 4-21.*, 2018.

[11] Y. Murthy and S. G. Koolagudi, "Content-based music information retrieval (cb-mir) and its applications toward the music industry: A review," *ACM Computing Surveys (CSUR)*, vol. 51, no. 3, p. 45, 2018.

[12] B. Logan *et al.*, "Mel frequency cepstral coefficients for music modeling.," in *ISMIR*, vol. 270, pp. 1–11, 2000.

[13] T. Lidy, C. N. Silla Jr, O. Cornelis, F. Gouyon, A. Rauber, C. A. Kaestner, and A. L. Koerich, "On the suitability of state-of-the-art music information retrieval methods for analyzing, categorizing and accessing non-western and ethnic music collections," *Signal Proc.*, vol. 90, 2010.

[14] G. Tsoumakas, I. Katakis, and I. Vlahavas, "A review of multi-label classification methods," in *Proceedings of the 2nd ADBIS workshop on data mining and knowledge discovery (ADMKD 2006)*, pp. 99–109, 2006.

[15] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proceedings of the IEEE conference on computer vision and pattern recognition*.

[16] A. Flexer, "A closer look on artist filters for musical genre classification," *World*, vol. 19, no. 122, pp. 16–17, 2007.

[18] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *2009 IEEE conference on computer vision and pattern recognition*, pp. 248–255, Ieee, 2009.

[17] Y. M. Costa, L. S. Oliveira, A. L. Koericb, and F. Gouyon, "Music genre recognition using spectrograms," in *2011 18th International Conference on Systems, Signals and Image Processing*, pp. 1–4, IEEE, 2011.

[19] X. Glorot, A. Bordes, and Y. Bengio, "Deep sparse rectifier neural networks," in *Proceedings of the fourteenth international conference on artificial intelligence and statistics*, pp. 315–323, 2011.

[20] Y. An, S. Sun, and S. Wang, "Naive Bayes classifiers for music emotion classification based on lyrics," in *IEEE/ACIS 16th Int. Conference on Computer and Information Science (ICIS)*, pp. 635–638, May 2017.

[21] A. M. Ujlambkar and V. Z. Attar, "Mood Classification of Indian Popular Music," in *Proc. of the CUBE International Information Technology Conference*, CUBE '12, (New York, NY, USA), pp. 278–283, ACM, 2012.

[22] R. Malheiro, R. Panda, P. Gomes, and R. P. Paiva, "Emotionally-Relevant Features for Classification and Regression of Music Lyrics," *IEEE Trans. on Affective Computing*, vol. 9, pp. 240–254, Apr. 2018.

[23] C. Laurier, J. Grivolla, and P. Herrera, "Multimodal Music Mood Classification Using Audio and Lyrics," in *2008 Seventh International Conference on Machine Learning and Applications*, pp. 688–693, Dec. 2008.

[24] X. Wang, Y. Wu, X. Chen, and D. Yang, "Enhance popular music emotion regression by importing structure information," in *Asia-Pacific Signal and Information Processing Association Annual Conference*, 2013.

[25] R. Mihalcea and C. Strapparava, "Lyrics, Music, and Emotions," in *Proceedings of the Joint Conference on Empirical Methods in Natural Language Processing and Computational Natural Language Learning*, (Jeju Island, Korea), Association for Computational Linguistics, 2012.

[26] C. Jareanpon, W. Kiatjindarat, T. Polhome, and K. Khongkraphan, "Automatic lyrics classification system using text mining technique," in *Int. Workshop on Advanced Image Technology (IWAIT)*, pp. 1–4, Jan. 2018.

[27] D. Yang and W. Lee, "Music Emotion Identification from Lyrics," in *11th IEEE International Symposium on Multimedia*, pp. 624–629, Dec. 2009.

[28] G. Tzanetakis and P. Cook, "Musical genre classification of audio signals," *IEEE Transactions on Speech and Audio Processing*.

[29] A. Berenzweig, B. Logan, D. P. Ellis, and B. Whitman, "A large-scale evaluation of acoustic and subjective music-similarity measures," *Computer Music Journal*, vol. 28, no. 2, pp. 63–76, 2004.

[30] P. Cano, E. Gomez, F. Gouyon, P. Herrera, M. Koppenberger, B. Ong, X. Serra, S. Streich, and N. Wack, "ISMIR 2004 audio description contest," tech. rep., Music Technology Group - Universitat Pompeu Fabra.

[31] I. Mierswa and K. Morik, "Automatic feature extraction for classifying audio data," *Machine Learning Journal*, vol. 58.

[32] F. Gouyon, A. Klapuri, S. Dixon, M. Alonso, G. Tzanetakis, C. Uhle, and P. Cano, "An experimental comparison of audio tempo induction algorithms," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 14, no. 5, pp. 1832–1844, 2006.

[33] D. Turnbull, L. Barrington, D. Torres, and G. Lanckriet, "Semantic annotation and retrieval of music and sound effects," *Trans. Audio, Speech and Lang. Proc.*, vol. 16, pp. 467–476, Feb. 2008.

[34] A. Holzapfel and Y. Stylianou, "Rhythmic similarity of music based on dynamic periodicity warping," in *Proceedings of The IEEE International Conf. on Acoustics Speech and Signal Processing*, pp. 2217–2220, 2008.

[35] K. Seyerlehner, G. Widmer, and P. Knees, "Frame level audio similarity-a codebook approach," in *Proceedings of The International Conference on Digital Audio Effects*, 2008.

[36] C. N. Silla Jr, A. L. Koerich, and C. A. Kaestner, "The latin music database.," in *Proceedings of The International Conference on Music Information Retrieval*, pp. 451–456, 2008.

[37] E. Law, K. West, M. I. Mandel, M. Bay, and J. S. Downie, "Evaluation of algorithms using games: The case of music tagging," in *Proceedings of the 10th International Society for Music Information Retrieval Conference, ISMIR 2009, Kobe Int. Conference Center, Kobe, Japan, October 26-30*.

[38] T. Bertin-Mahieux, D. P. Ellis, B. Whitman, and P. Lamere, "The million song dataset.," in *Proceedings of The International Conference on Music Information Retrieval*, vol. 2, p. 10, 2011.

[39] M. Goto, H. Hashiguchi, T. Nishimura, and R. Oka, "Rwc music database: Popular, classical and jazz music databases.," in *Proc. of The International Conference on Music Information Retrieval*, vol. 2, pp. 287–288, 2002.